Saihritesh Kowtarapu

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Summary

I am a CS undergraduate and AI/ML Engineering Intern who builds end-to-end Generative AI systems (RAG pipelines, agentic tool-calling workflows, fine-tuned LLMs) that run in production. I bring strong foundations in Data Structures & Algorithms and System Design, and experience presenting technical results to engineering leadership.

Education

PES University — Bengaluru, India
Bachelor of Technology in Computer Science Engineering; CGPA: 8.59/10 — *Sep 2023 – June 2027 (Expected)*

Experience

EverestIMS Technologies — Jul 2026 – Sep 2026
AI/ML Engineer Intern — *Bengaluru, India*

- Built and deployed a production RAG + database-routing assistant for Airtel's network operations team, now in use by the client. (see Projects)
- Fine-tuned three 7–8B open-source LLMs with QLoRA for network root-cause analysis on a leakage-free dataset built from a 304-entry knowledge base, reaching 95% accuracy and 0.93 macro-F1 on a held-out benchmark.
- Designed an LLM-as-judge evaluation pipeline and ran controlled experiments on context length, noise ratio and evidence position to measure how reliably each model grounded its answers in supplied evidence.
- Presented a 40-slide model selection playbook to engineering leadership, covering evaluation results, model trade-offs and recommendations for private on-premise deployment.

Projects

Airtel Network Intelligence Assistant | *Python, MongoDB, ChromaDB, Ollama, Groq* — Jul 2026 – Sep 2026

- Built a chat assistant that routes each operations question to a live MongoDB database (∼600K records) or a RAG pipeline over 15 technical documents, answering strictly from retrieved data
- Designed an embedding-based query router reaching 93% accuracy on an adversarial set of ambiguous questions; traced misrouting to class imbalance and a sampling gap, then fixed it with per-class scoring
- Added a validation layer that blocks unsafe LLM-generated queries and cross-checks aggregations against raw counts; deployed on a Linux server and adopted by Airtel

Notoroma, AI-Integrated Notes Platform | *Django, NextJS, Groq* | notoroma.vercel.app — Sep 2026

- Built an agentic "Study Buddy" tutor as a custom multi-round tool-calling loop on Groq's function-calling API, using a two-phase prompt that separates tool selection from JSON output
- Implemented a chunk-and-map-reduce pipeline for length-agnostic summarization, flashcard and test generation, run asynchronously through Celery/Redis to keep long LLM calls off the request path
- Designed a normalized PostgreSQL schema and full Django CRUD for multi-user accounts and markdown notes

E-Commerce Business Analytics & Forecasting | *Python, Pandas, SQL, Tableau* — Jun 2026 – Jul 2026

- Integrated 8 relational datasets (100K+ orders) into a unified analytical model, resolving merge conflicts and duplicate records
- Engineered features such as delivery delay, customer cohorts and repeat-purchase flags, then computed 5 KPIs covering revenue forecasting and seller performance, visualized in a Tableau dashboard

Open-Source Contributions, Pandas | *Python, Pandas, Pytest, Git* — Jul 2026 – Oct 2026

- Fixed a legend-rendering defect in DataFrameGroupBy.plot.scatter() by tracing execution through core Pandas internals, with targeted Pytest coverage and zero regressions across the existing suite(PR#66027).
- Fixed a 2.x regression where DataFrame.ffill/bfill with axis=1, inplace=True silently left the original unchanged; traced it to an early return in the transposed branch of NDFrame._pad_or_backfill and added a parametrized regression test which has been merged(PR#71232).

Technical Skills

Generative AI: RAG, Agentic Workflows, Tool/Function Calling, LangChain, LangGraph, Prompt Engineering, LLM Evaluation (LLM-as-judge), Fine-Tuning (LoRA/QLoRA), Hugging Face
ML & Data: PyTorch, TensorFlow, Pandas, NumPy, Statsmodels, Scikit-learn, SQL, Tableau
Backend & Infra: FastAPI, Django, Flask, Celery, Redis, PostgreSQL, MongoDB, Vector DBs (ChromaDB, Qdrant, pgvector), Google Cloud Platform, Linux
Languages & Fundamentals: Python, SQL, C, JavaScript; Data Structures & Algorithms, System Design, REST API Design, Pytest, Git

Certifications

Building AI Agents with MongoDB | *MongoDB* Jul 2026
Introduction to TensorFlow for AI, ML and Deep Learning | *Stanford University* Jul 2025
Convolutional Neural Networks in TensorFlow | *Stanford University* Jul 2025
Supervised Machine Learning: Regression | *Stanford University* Mar 2025
HTML, CSS, JavaScript | *Meta* Jul 2024